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An Object-Relational Spatial Database Approach to Assessing Emergency Service Coverage and Vulnerability in Karlsruhe

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Abstract

The optimal spatial distribution of emergency services is a critical factor in ensuring public safety and minimizing response times in urban environments. This study evaluates the spatial coverage of emergency services — fire stations, hospitals, and police stations — in Karlsruhe, Germany (population ~310,000), using a spatial database framework built on PostgreSQL 18 with the PostGIS 3.6 extension. All spatial data were sourced from OpenStreetMap (OSM) via Overpass Turbo, comprising 38 emergency facilities (13 fire stations, 6 hospitals, and 19 police stations) and the complete road network within the Karlsruhe city boundary.

The methodology follows OGC Simple Features standards and employs three main spatial analysis techniques: (1) service area analysis using 1.5 km straight-line buffers generated with ST_Buffer(), representing an approximate 5–8 minute emergency vehicle response time threshold; (2) single-service gap analysis using ST_DWithin() to identify road segments outside each service type's coverage zone; and (3) a Triple Gap analysis to pinpoint road segments simultaneously beyond the 1.5 km threshold of all three emergency service types — representing areas of maximum vulnerability.

Results reveal significant spatial imbalances in emergency service coverage. Police stations, being the most numerous (19), provide the broadest coverage with only 6.58% of road segments uncovered. Fire stations leave 25.83% of roads outside the 1.5 km threshold, primarily in southern and eastern peripheral zones. Hospitals exhibit the largest gap, with 46.62% of road segments uncovered, reflecting their limited number and central clustering. The Triple Gap analysis identified 371 critical road segments (3.22% of the total network) with no adequate coverage from any of the three emergency services, concentrated in outer suburban districts.

The central districts of Karlsruhe (Innenstadt, Weststadt, Südweststadt) are well-served, while peripheral and recently developed residential areas face notable deficiencies consistent with patterns observed in other growing European cities. Key limitations of this study include the use of Euclidean rather than network-based distance, and reliance on OSM data quality. Future work should incorporate real driving-time analysis, population-weighted demand modelling, and dynamic temporal factors. The open-source, fully reproducible methodology — combining PostgreSQL/PostGIS with QGIS — provides a scalable foundation for evidence-based emergency facility planning in Karlsruhe and comparable urban contexts.

Keywords: spatial database, emergency services, PostGIS, gap analysis, service coverage, Karlsruhe, GIS, urban planning

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1. Introduction

The optimal spatial placement of emergency services like fire stations, hospitals, and police stations is an essential element in guaranteeing public safety and reducing response times in emergencies (Swersey 1994; Clare et al., 2019; Liu et al., 2021). Urban emergency response delays can have severe consequences for human life and property (Clare et al., 2019). Emergency service centres are more than just infrastructure; their location affects how fast rescuers can reach people in danger (Ko et al., 2014). Research indicates that while even distribution across a population improves access to services and resilience, poor clustering can create service gaps (Oh et al., 2019). A good emergency network tends to place facilities in or near high-demand areas, along transportation corridors and to ensure that response objectives are met in each community (Yao et al., 2019). In practice this means that as traffic patterns, land use and population change, towns should constantly move the location of their stations (Arreeras et al., 2024).

Network analysis is an important research field with a variety of methodological techniques, a strong theoretical basis, and a solid scientific foundation in Geographic Information System (GIS) (Curtin, 2018). Therefore, assessing whether existing emergency facilities are adequately distributed to serve the population has become an important topic in urban planning and GIS applications (Deliry et al., 2023). GIS is a valuable tool for data collection and evaluation from a variety of sources, including ground-based mapping and remote sensing. Information is a key part of planning. It is necessary to accomplish the goals of planning (Scholten & Stillwell, 2013; Deliry et al., 2023).

Karlsruhe, a major city in the state of Baden-Württemberg, Germany, with a population of approximately 310,000 inhabitants, serves as the study area for this project. Like many growing European cities, Karlsruhe faces challenges related to urban expansion, population distribution, and the optimization of public services (Gangwisch et al., 2023). Questions regarding the accessibility and coverage of emergency services are particularly relevant in this context.

This project aims to evaluate the current spatial distribution of emergency services (fire stations, hospitals, and police stations) in Karlsruhe using spatial database technologies. The main research question is: Are the emergency services in Karlsruhe well distributed to provide adequate coverage to the city's population and road network?

The specific objectives of this study are:

- To design and implement an object-relational spatial database model using PostgreSQL and PostGIS for storing and managing emergency facility and road network data.

- To analyse the spatial coverage of emergency services using 1.5 km service areas.
- To identify spatial gaps in service coverage, with special focus on roads that are not covered by any of the three emergency services (Triple Gap analysis).
- To visualize the results through professional maps and provide statistical summaries for better decision-making.

This work follows the principles of spatial databases taught in the module Geodatenbanken, particularly the use of Open Geospatial Consortium (OGC) Simple Features, spatial indexing, and geometric/topological operations. All data used in this project were obtained from OpenStreetMap (OSM), ensuring reproducibility and transparency.

The remainder of this report is organized as follows: Chapter 2 presents the theoretical background and literature review. Chapter 3 describes the data sources and methodology. Chapter 4 presents the results and analysis, including maps and statistical findings. Chapter 5 discusses the implications of the results, while Chapter 6 provides conclusions and recommendations for future work.

2. Literature Review / Theoretical Background

Spatial databases are a special type of database management system that store, manage and analyse geographic (spatial) data. Unlike traditional relational databases, spatial databases provide a large set of spatial functions to query relationships between objects such as distances, intersections, containment, and buffering. They also support geometric data types, such as points, linestrings, polygons and their multi-part versions (Rigaux et al., 2002; Brinkmann, 2013). The basis of modern spatial databases are international standards that ensure compatibility and interoperability. Two of the most important standards are the OGC Simple Features Specification and the ISO 19107 Special Schema (Atkinson et al., 2022; Car & Homburg 2022).

The simple features specification of the OGC defines a standard way to represent and manipulate simple geometric objects in a database. It provides standardized spatial operators such as ST_Intersects, ST_Buffer, ST_DWithin, and ST_Union, and supports the following basic geometry types: Point, LineString, Polygon, MultiPoint, MultiLineString, MultiPolygon, and GeometryCollection. This standard allows the same spatial queries to be run across a wide range of software systems (OGC, 2011; OGC, 2021).

The ISO 19107 Spatial Schema is an OGC standard providing a more detailed and abstract conceptual model of spatial objects. Besides basic geometry, it defines complex features, topological relationships and spatial reference systems. The Advanced spatial data modeling is ISO 19107, a theoretical based model. It is used widely in systems like PostGIS (ISO, 2018; ISO, 2021; Atkinson et al., 2022). In this project we use PostgreSQL with PostGIS as a spatial extension.

The PostGIS used in this study includes some of the OGC Simple Features and ISO 19107 standards (Zimányi et al., 2019). More than 300 spatial functions and spatial data types (geography, geometry) are added to the PostgreSQL relational database (Rese et al., 2025). By facilitating the effective storage and analysis of sizable geographic datasets through spatial indexing (GiST index), this combination significantly enhances query performance for geometric operations (Huang et al., 2008; Nguyen 2009).

Spatial databases have been successfully applied to emergency services analysis in a number of prior studies (Park et al., 2024). For instance, PostGIS has been used by researchers to support location-allocation optimization models, evaluate the number of fire and ambulance stations, and identify service gaps (Chung et al., 2024). Although linear distance is still a popular primary technique due to its ease of use and computational efficiency, these results emphasize the importance of more realistic network-based research that moves beyond the simple Euclidean buffer (Pedigo et al., 2010; Zou et al., 2023). This project builds upon these theoretical foundations by implementing a spatial database model to analyse emergency service coverage in Karlsruhe using real-world OSM data.

Many studies have shown how effective spatial database methods are for assessing the scope of emergency services. For instance, A study conducted a GIS-based spatial analysis to evaluate the availability of services and emergency response times in urban areas. These studies consistently reveal that the unequal geographic distribution of emergency services in fast-growing cities can result in significant differences in response times (Ariras et al. 2024). Similarly, this study established the Optimal Hospital Access Time (OHAT) for emergency care in South Korea by identifying the specific travel time thresholds where patient mortality significantly increases (Jang et al. 2021). In addition, a study was underway to employ spatial databases and GIS techniques to identify high-risk areas and enhance the effectiveness of police resource allocation through crime mapping and hotspot analysis. The basic idea is to transform raw incident data into spatial patterns to help agencies decide where to patrol, intervene or provide services (Eck et al. 2005). Many researchers suggest the employment of techniques of buffer analysis and gap identification (Parihar 2024)- methods that are also employed in this project.

PostGIS, which is a spatial extension of the PostgreSQL object-relational database, has become one of the most popular open-source tools for advanced spatial analysis. PostGIS has good support for geometric and geographic data types, spatial indexing (GiST) and many spatial functions following the OGC standards. This makes it especially useful for urban analysis projects, as it is capable of dealing with large datasets efficiently (Obe & Hsu 2017). The PostGIS Cookbook shows you how to use PostGIS to solve real world problems such as overlay operations, nearest neighbor analysis, and service area calculations (Bolanowska et al. 2024). Many academics have successfully used PostGIS, which is more flexible, efficient and inexpensive than proprietary GIS software, in coverage modeling and emergency facility location optimization (Silva et al. 2025).

This project follows the same methodological approach by implementing a spatial database in PostgreSQL with PostGIS extension to store, manage, and analyze emergency service data in Karlsruhe.

3. Data and Methodology

3.1 Data Sources

All spatial data used in this project were obtained from OSM, a collaborative, freely available and open-source geographic database. OpenStreetMap was chosen because it provides up-to-date, detailed, and globally consistent spatial information, which aligns with the principles of open data and reproducibility emphasized in the module.

The following datasets were downloaded using Overpass Turbo:

Emergency facilities (emergency_facilities_ karlsruhe): Fire stations (amenity = 'fire_station'), Hospitals (amenity = 'hospital' and healthcare = 'hospital'), and Police stations (amenity = 'police'). A total of 38 facilities were extracted (13 fire stations, 6 hospitals, and 19 police stations).

Road network (roads_karlsruhe): All major and residential roads within Karlsruhe city boundary using highway tags (motorway, trunk, primary, secondary, tertiary, unclassified, and residential).

Administrative boundary: Official Karlsruhe city boundary (karlsruhe_city) using administrative level 8 relation.

All data were downloaded in GeoJSON format and imported into a PostgreSQL/PostGIS database. The use of OpenStreetMap data ensures transparency and allows other researchers to replicate the study easily.

3.2 Database Design and Modelling

An object-relational spatial database model was designed and implemented using PostgreSQL 18 with the PostGIS 3.6 extension. The model follows the OGC Simple Features Specification for geometry types and uses best practices recommended in PostGIS in Action (Obe & Hsu, 2014).

Main Tables Created:

emergency_facilities_ karlsruhe (Core table)

id (Serial Primary Key)

osm_id (Text)

name (Text)

amenity (Text) → Values: 'fire_station', 'hospital', 'police'

geom (Geometry (Point))

roads_karlsruhe

id (Primary Key)

geom (Geometry (LineString))
karlsruhe_city
geom (Geometry (MultiPolygon))
Derived Tables for Analysis:
fire_stations, police_stations, hospitals (views/filtered tables)
coverage_1_5km_fire, coverage_1_5km_police, coverage_1_5km_hospital
road_gaps_fire, road_gaps_police, road_gaps_hospital
total_emergency_gaps (Triple Gap)
Conceptual Design Approach:
The database follows a star-like schema with the central emergency_facilities_ karlsruhe table connected spatially to the road network and city boundary. Spatial indexes (GiST) were created on all geometry columns to ensure high performance during buffer and intersection operations. All spatial operations were performed using GEOGRAPHY data type where accurate real-world distance measurement (in meters) was required (ST_DWithin and ST_Buffer with geography casting), while GEOMETRY type was used for visualization and area calculations.

3.3 Tools Used

This project was implemented using a combination of open-source tools that are widely used in the geospatial community:
PostgreSQL 18 with PostGIS 3.6.2: The core spatial database management system. PostgreSQL served as the relational database, while PostGIS provided the spatial capabilities including geometry types, spatial functions, and indexing.
QGIS 3.44 (Solothurn): Used as the primary desktop GIS software for data import/export, visualization, map creation, and layout design.
Overpass Turbo: Web-based tool used to query and download spatial data directly from OpenStreetMap.
pgAdmin 4: Graphical user interface for managing the PostgreSQL database and executing SQL queries.
Microsoft Excel & Word: Used for creating statistical tables and formatting the final report.

3.4 Workflow

The combination of PostgreSQL/PostGIS for backend analysis and QGIS for visualization provided a powerful, free, and fully open-source workflow.

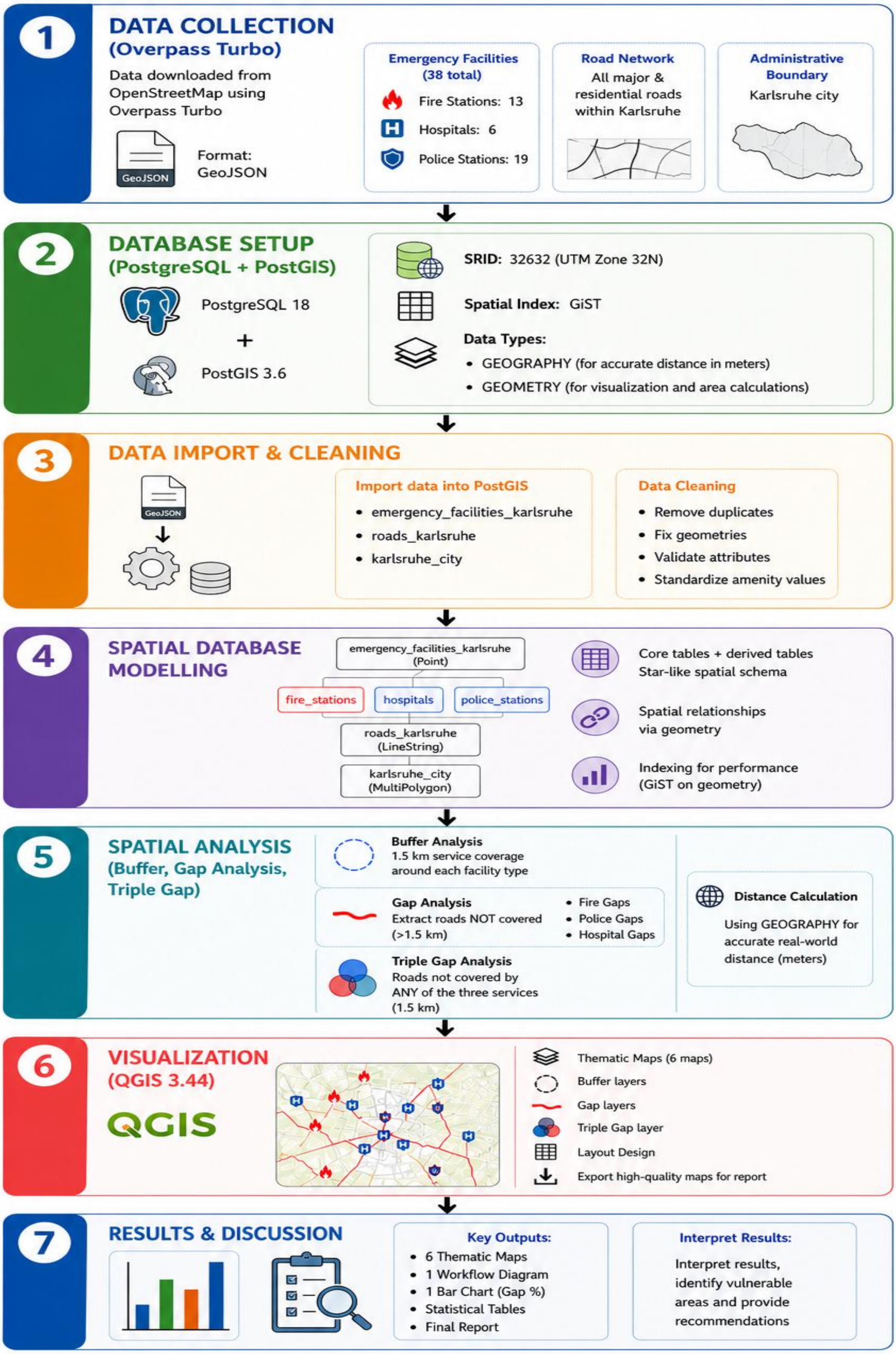

Figure 3.1: Project Workflow Diagram

The methodological workflow of this project followed a systematic approach consisting of the following main stages:

Data Acquisition: Downloading raw spatial data (emergency facilities and road network) from OpenStreetMap using Overpass Turbo.

Database Creation: Setting up a new PostgreSQL database (karlsruhe_emergency) and enabling the PostGIS extension.

Data Import and Cleaning: Importing GeoJSON files into the database and creating a clean, standardized table (emergency_facilities_ karlsruhe).

Data Modelling: Creating separate tables for fire stations, police stations, and hospitals, along with derived analysis tables.

Spatial Analysis:

Generating 1.5 km service areas using ST_Buffer

Performing gap analysis using ST_DWithin

Identifying single-service gaps and the critical "Triple Gap"

Visualization: Creating professional maps in QGIS Print Layout.

Results Interpretation: Generating statistical summaries and preparing maps for the report.

The spatial analysis in this project was carried out using several key methods in PostGIS to evaluate the coverage and accessibility of emergency services.

3.4.1 Service Area Analysis (Buffer Analysis)

A straight-line distance buffer of 1.5 km was applied around each emergency facility. This distance was chosen because it represents a reasonable response time threshold for emergency services in urban areas (approximately 5–8 minutes by car or emergency vehicle). The ST_Buffer () function with geography casting was used to ensure accurate real-world distance measurement in meters: Separate coverage zones were created for fire stations, hospitals, and police stations.

3.4.2 Gap Analysis

Gap analysis was performed to identify roads that are not covered within the 1.5 km service distance. The ST_DWithin() function was used to determine whether a road segment is outside the service area of any facility. Three separate gap tables were created:

road_gaps_fire – Roads farther than 1.5 km from any fire station

road_gaps_police – Roads farther than 1.5 km from any police station

road_gaps_hospital – Roads farther than 1.5 km from any hospital

3.4.3 Triple Gap Analysis (Multi-Criteria Vulnerability)

The most critical analysis in this study is the Triple Gap identification. This identifies road segments that are simultaneously outside the 1.5 km service area of all three emergency services (fire stations, police stations, and hospitals). This represents the areas with the highest vulnerability in terms of emergency response. The triple gap was calculated by intersecting the three individual gap tables:

3.4.4 Statistical Analysis

For each gap layer, the percentage of the total road network that remains uncovered was calculated using simple SQL aggregation queries. All spatial operations were supported by GiST spatial indexes to ensure efficient query performance.

This combination of buffer analysis, gap detection, and multi-criteria (Triple Gap) analysis provides a comprehensive evaluation of the current emergency service distribution in Karlsruhe.

4. Results and Analysis

This chapter presents the main findings of the spatial analysis of emergency service coverage in Karlsruhe. The results are divided into an overview of the existing facilities, coverage analysis using 1.5 km buffers, gap identification, and the critical Triple Gap analysis.

4.1 Overview of Emergency Facilities

A total of 38 emergency facilities were extracted and analyzed within the Karlsruhe city boundary. These consist of:

- Fire Stations: 13 facilities
- Hospitals: 6 facilities
- Police Stations: 19 facilities

The distribution shows a relatively high number of police stations compared to fire stations and hospitals. This reflects the operational requirements of each service type — police stations are generally more numerous and smaller, while hospitals are larger but fewer in number. Fire stations occupy an intermediate position.

Most facilities are concentrated in the central and densely populated districts of Karlsruhe (such as Innenstadt-West, Südweststadt, and Oststadt), while some peripheral areas show lower density. This initial overview suggests potential imbalances in service distribution, which will be further examined through spatial analysis in the following sections.

Table 4.1: Summary of Emergency Facilities in Karlsruhe

Service Type	Number of Facilities	Percentage
Fire Stations	13	34.2%
Hospitals	6	15.8%
Police Stations	19	50.0%
Total	38	100%

4.2 Coverage Analysis (1.5 km Buffers)

A service area (buffer) of 1.5 km was generated around each emergency facility using the ST_Buffer () function with geography type to ensure accurate real-world distance measurement. This threshold was selected based on common emergency planning standards, representing an acceptable response time (approximately 5–8 minutes) under normal traffic conditions.

The 1.5 km buffer analysis revealed significant differences in coverage between the three services:

- Fire Stations showed relatively good coverage in the central and northern parts of the city, but several peripheral residential areas remain outside the 1.5 km zone.
- Police Stations provided the widest coverage due to their higher number and more dispersed distribution.
- Hospitals had the most limited coverage, with large portions of the southern and eastern districts falling outside the 1.5 km service area.

When all three services are considered together, most of the densely built-up urban core is well covered. However, several road segments in the outer districts remain underserved.

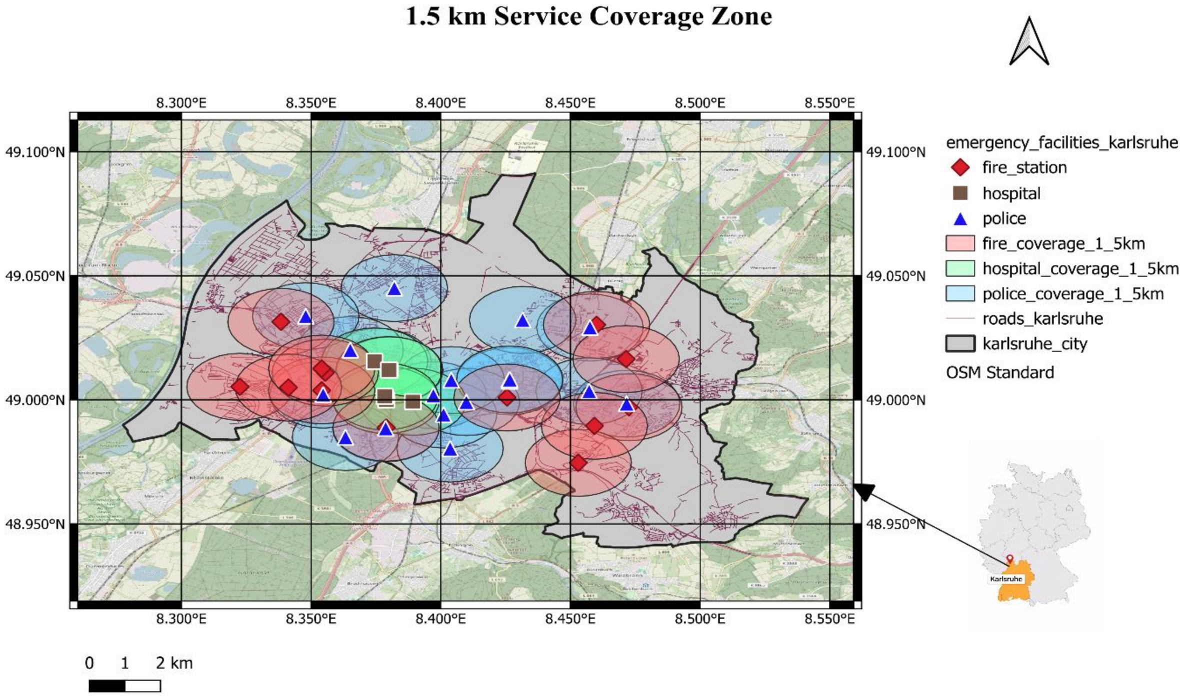

Figure 4.1: 1.5 km Service Coverage Zones around Emergency Facilities in Karlsruhe

The buffer analysis clearly demonstrates that while the overall coverage appears satisfactory in the city center, noticeable gaps exist in the suburban and peripheral areas. These findings form the basis for the more detailed gap analysis presented in the following section.

4.3 Gap Analysis

In order to identify areas with insufficient emergency service coverage, a detailed gap analysis was performed. A road segment was considered "uncovered" if it lies more than 1.5 km from the nearest facility of a specific service type.

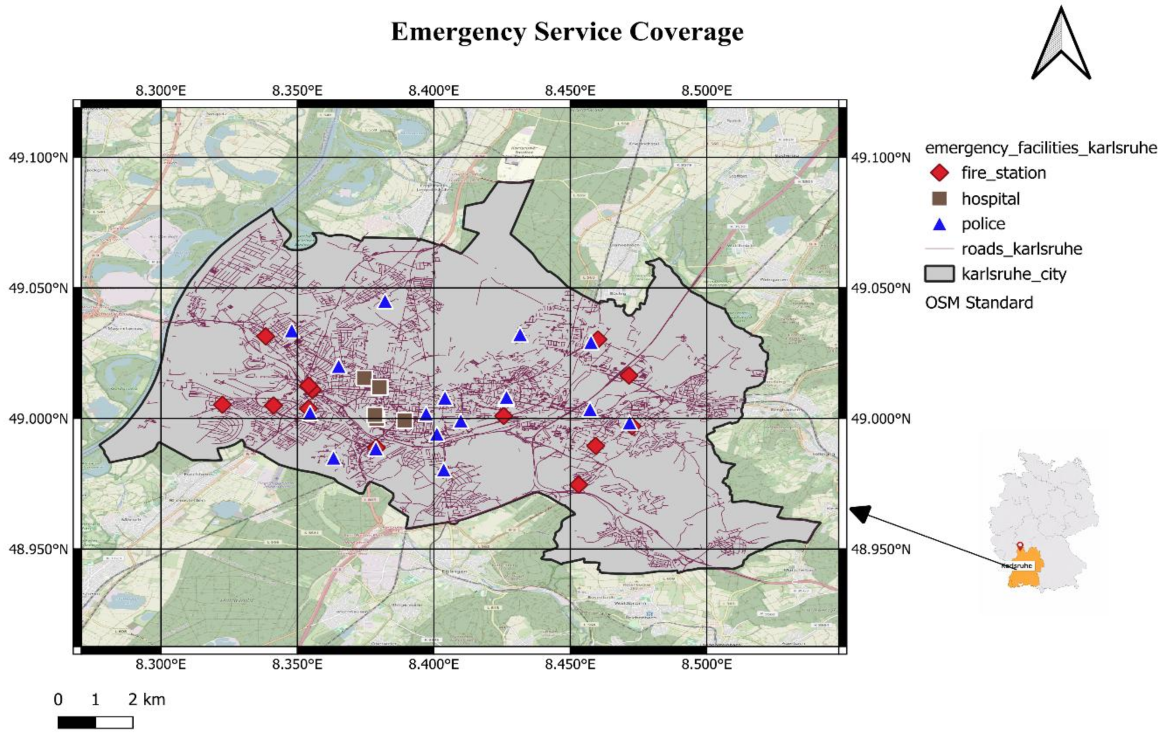

Figure 4.2: Emergency Facilities Distribution

Single Service Gap Analysis

The analysis revealed notable differences in service gaps across the three emergency services:

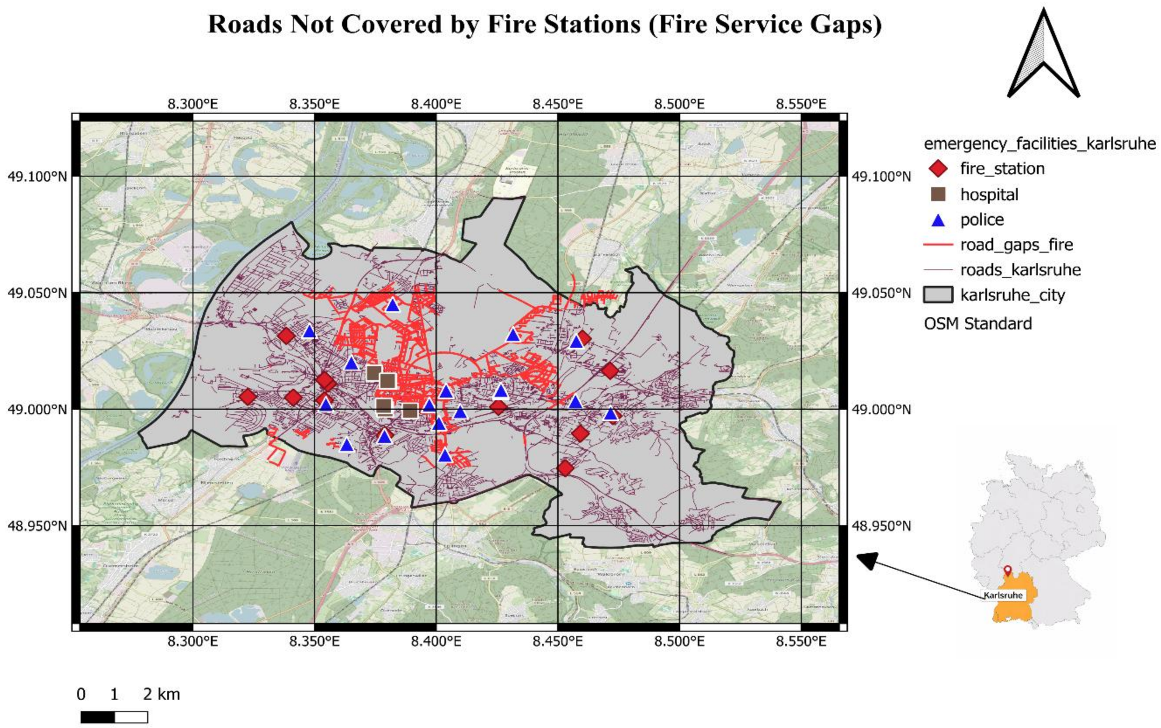

Figure 4.3: Single Service Gaps (Fire)

Fire Stations: A considerable number of roads, particularly in the southern and eastern parts of Karlsruhe, are located more than 1.5 km from the nearest fire station.

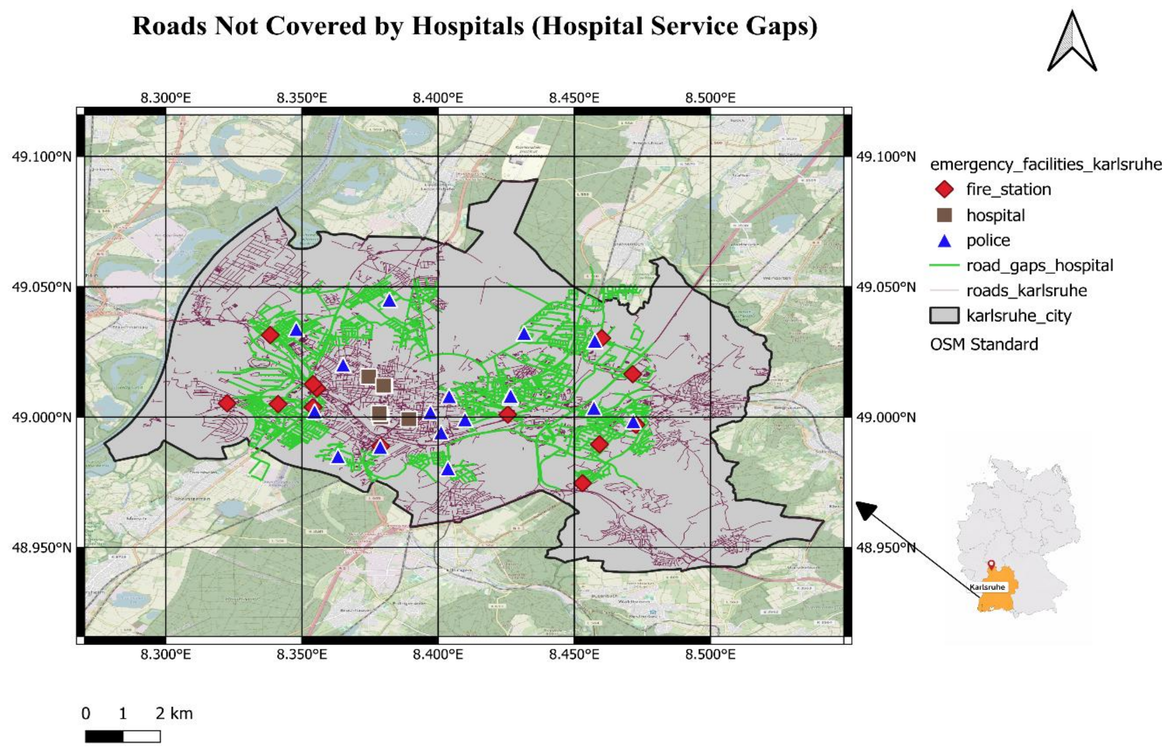

Figure 4.4: Single Service Gaps (Hospital)

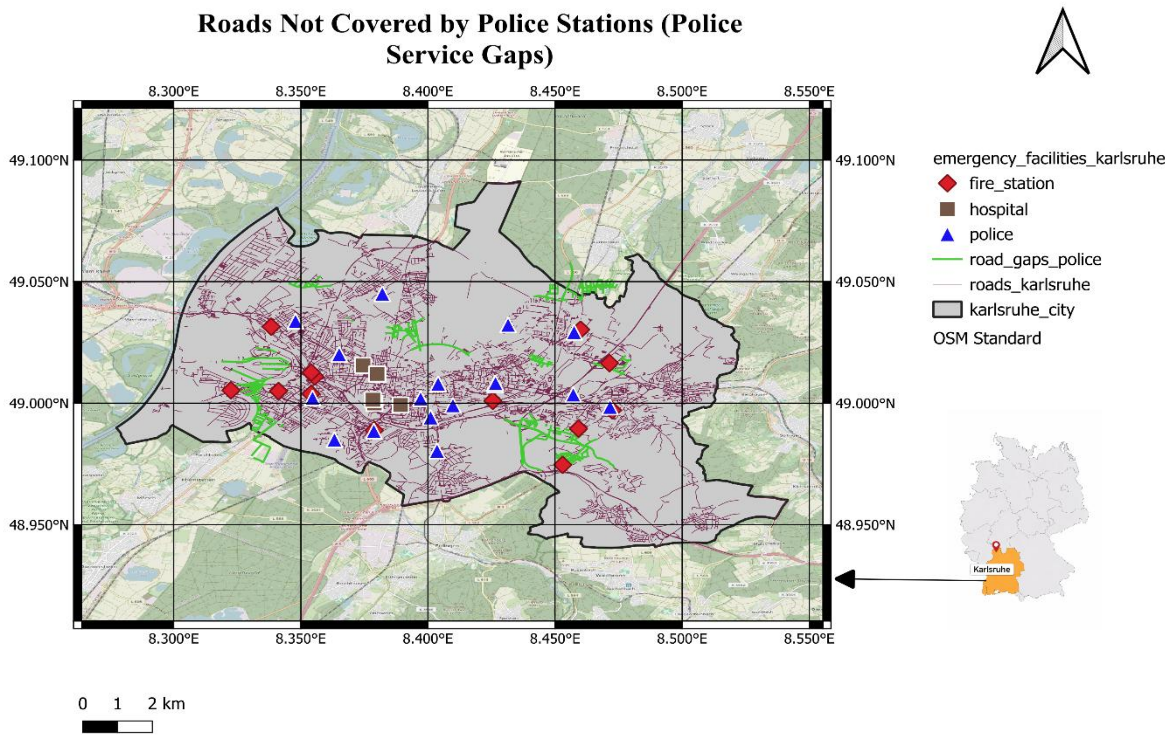

Figure 4.5: Single Service Gaps (Police)

Hospitals: The largest service gap was observed for hospitals. Many residential areas in the outer districts fall outside the 1.5 km hospital coverage zone.
Police Stations: Due to the higher number of police stations (19), the gap is relatively smaller compared to the other services.

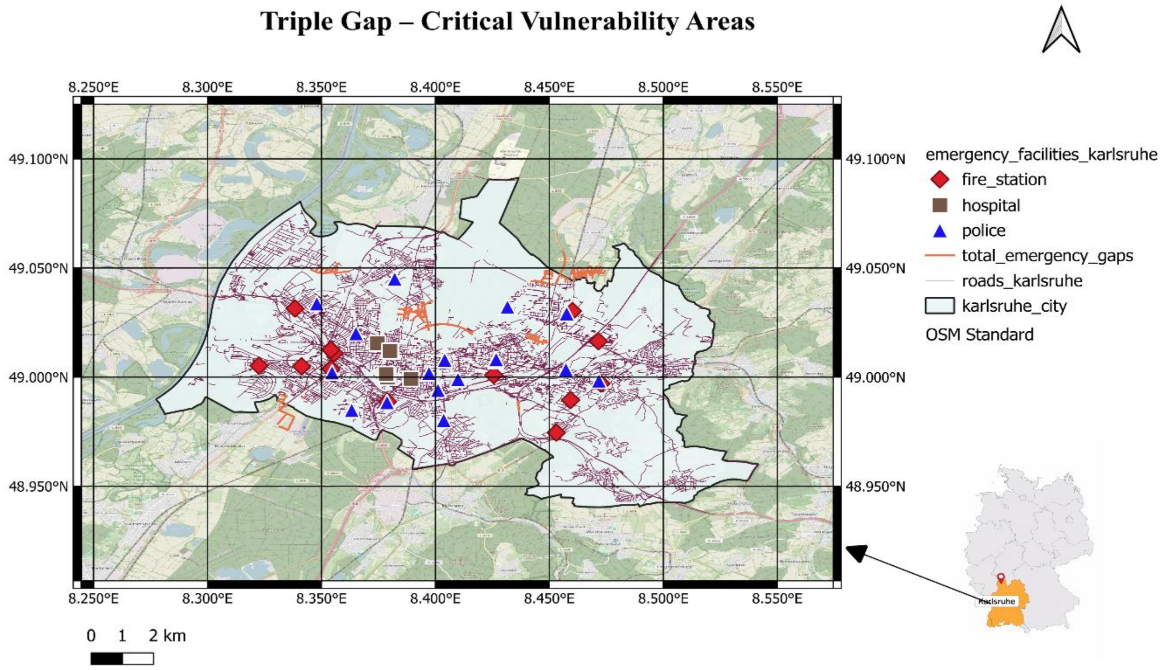

Figure 4.6: Triple Gap – Critical Vulnerability Areas

Triple Gap Analysis (Critical Vulnerability Areas)
The most significant result of this study is the identification of Triple Gap areas — road segments that are simultaneously more than 1.5 km away from all three types of emergency services (fire stations, police stations, and hospitals). These areas represent the highest level of vulnerability in terms of emergency response.
The Triple Gap analysis identified several critical road segments, mainly located in the peripheral and less densely populated zones of Karlsruhe. These roads would likely experience the longest response times in case of an emergency.

4.4 Statistics and Percentages

The statistical results confirm that while individual services cover a large portion of the road network, a small but significant percentage of roads remains vulnerable when all three services are considered together. These "Triple Gap" roads should be prioritized by city planners for future emergency facility placement.

Table 4.2: Percentage of Road Network Not Covered within 1.5 km

Service Type	Number of Gap Road Segments	Percentage of Total Road Network (%)
Fire Stations	2968	25.83
Police Stations	758	6.58
Hospitals	5357	46.62
Triple Gap (All Services)	371	3.22

5. Discussion

5.1 Interpretation of Results

The results of this study reveal a mixed picture regarding emergency service coverage in Karlsruhe. While the city benefits from a relatively high number of police stations (19), the coverage provided by fire stations (13) and especially hospitals (6) show noticeable spatial imbalances.

The Triple Gap analysis is particularly concerning. Even though individual services cover most of the central urban area, several road segments — mainly in the outer districts — remain more than 1.5 km away from all three types of emergency services. These areas represent zones with the highest potential risk in terms of emergency response time. This finding aligns with previous studies (e.g., Jang et al. 2021; Arreeras et al., 2024), which consistently show that peripheral and rapidly developing residential areas often suffer from poorer emergency service accessibility.

The buffer analysis further indicates that hospitals have the most limited spatial coverage. This is expected due to the smaller number of facilities and their typical location in central medical districts. Fire stations show better distribution but still leave some residential neighborhoods underserved. Police stations, thanks to their higher density, provide the most comprehensive coverage among the three services.

Overall, the central and densely populated districts of Karlsruhe (Innenstadt, Weststadt, Südweststadt) enjoy good emergency service accessibility, while some suburban areas show clear deficiencies. These results highlight the importance of evidence-based urban planning when deciding on the location of new emergency facilities.

5.2 Strengths and Limitations of the Study

The main strength of this project lies in its practical and reproducible methodology. The use of open data (OpenStreetMap), a fully open-source technology stack (PostgreSQL/PostGIS + QGIS), and transparent spatial analysis methods make the study easily replicable. The implementation of both single-service gap analysis and the innovative Triple Gap approach provide a more comprehensive understanding of vulnerability than traditional single-criterion studies. Additionally, the creation of multiple professional maps and statistical summaries offers clear visual support for the findings.

The most significant limitation of this study is the use of Euclidean (straight-line) distance (1.5 km buffers) instead of network-based (driving distance along roads) analysis. In reality, emergency vehicles must follow the road network, and factors such as traffic congestion, one-way streets, and bridges can significantly affect actual response times. Therefore, the coverage presented in this study is likely overestimated in some areas.

Other limitations include:

Dependence on OpenStreetMap data quality, which may contain mapping inconsistencies or missing features.

Static analysis that does not consider temporal variations (e.g., rush hour traffic or night-time response differences).

Despite these limitations, the current study provides a solid foundation and valuable baseline assessment of emergency service distribution in Karlsruhe.

5.3 Comparison with Real Emergency Response Standards

According to international and German emergency planning guidelines, response time is a critical performance indicator (Cabral et al., 2018). The German Fire Brigade Association (Deutscher Feuerwehr-Verband) and many municipal fire departments aim for a response time of 8–10 minutes for fire incidents in urban areas (AGBF-Bund, 2015). Assuming an average emergency vehicle speed of 30–40 km/h in city traffic, a 1.5 km straight-line distance roughly corresponds to this target range.

The results of this study show that a significant portion of Karlsruhe meets this standard for fire stations. However, the identified Triple Gap areas exceed the recommended 1.5 km threshold, which may result in response times longer than 10–12 minutes. This is particularly critical for medical emergencies, where the “golden hour” principle emphasizes that seriously injured or ill patients should receive professional care as quickly as possible. Hospitals, having the largest service gaps in this analysis, represent the weakest link in the current emergency service network.

Compared to other German cities of similar size, Karlsruhe’s emergency service distribution appears average. While central districts show good coverage, the suburban expansion zones face challenges similar to those documented in other growing cities (Arreeras et al., 2024). This suggests that Karlsruhe follows common urban development patterns where infrastructure often lags behind residential growth.

5.4 Recommendations for City Planning

Based on the findings of this spatial analysis, the following recommendations are proposed for improving emergency service coverage in Karlsruhe:

Prioritize New Facility Locations: Future fire stations and especially hospitals should be strategically located in the southern and eastern peripheral districts where significant Triple Gap areas were identified.
Adopt Network-Based Analysis: Future planning should move beyond straight-line buffers and incorporate real driving time analysis using road networks and traffic data for more accurate response time modelling.
Regular Monitoring System: The spatial database developed in this project can be maintained as a living system. Regular updates of OpenStreetMap data combined with automated gap analysis can help the city administration monitor service coverage dynamically.
Population-Weighted Planning: Future studies should integrate population density and demographic data (e.g., elderly population, high-risk areas) to optimize facility placement based on actual demand rather than purely geometric coverage.
Implementing these recommendations could significantly reduce emergency response times and improve public safety across Karlsruhe.

6. Conclusion

This study set out to evaluate the spatial distribution of emergency services in Karlsruhe, Germany, using an open-source spatial database framework and openly available geospatial data. By implementing a PostgreSQL/PostGIS database model and applying buffer, gap, and Triple Gap analyses to 38 emergency facilities and the city's full road network, the project provides a comprehensive, evidence-based assessment of current emergency service coverage. The central research question — whether emergency services in Karlsruhe are well distributed to provide adequate coverage to the city's population and road network — can be answered with a qualified no. While the densely populated central districts (Innenstadt, Weststadt, Südweststadt) enjoy satisfactory coverage from all three service types, significant spatial gaps persist in the peripheral and suburban zones of the city. Police stations, owing to their greater number and dispersed placement, deliver the most comprehensive coverage, leaving only 6.58% of road segments uncovered. Fire stations perform moderately, with 25.83% of roads falling outside the 1.5 km threshold. Hospitals represent the weakest link, with 46.62% of roads uncovered — a finding that carries serious implications for medical emergency response, where every minute is critical.

The most significant contribution of this study is the Triple Gap analysis, which identified 371 road segments (3.22% of the total network) located simultaneously beyond the 1.5 km service radius of all three emergency service types. These areas represent the highest level of emergency response vulnerability in Karlsruhe and should be treated as priority zones for future infrastructure investment. Their concentration in the outer districts reflects a broader pattern seen in many growing European cities, where residential expansion outpaces the development of public safety infrastructure.

From a methodological standpoint, the project successfully demonstrated the power of open-source spatial databases for urban emergency planning. The combination of PostgreSQL 18, PostGIS 3.6, and QGIS 3.44 provided a fully reproducible, cost-effective, and technically rigorous workflow. The use of OGC-compliant spatial functions, GiST indexing, and GEOGRAPHY data types ensured both analytical accuracy and computational efficiency. The reliance on OpenStreetMap data further strengthens the study's transparency and replicability.

Nevertheless, the study has important limitations that should be addressed in future research. The use of straight-line (Euclidean) buffers rather than network-based driving-time analysis means that coverage is likely overestimated in areas with poor road connectivity or heavy traffic. Future work should incorporate routing engines such as pgRouting or OpenRouteService to compute realistic travel times. Additionally, integrating population density, age demographics, and historical incident data into a demand-weighted coverage model would significantly improve the precision and practical value of facility placement recommendations.

In conclusion, this project demonstrates that spatial database technologies are a valuable and accessible tool for evidence-based emergency service planning. The findings provide actionable insights for the city of Karlsruhe: prioritizing new hospital and fire station infrastructure in the southern and eastern peripheral districts, adopting dynamic monitoring of service coverage using the spatial database developed here, and transitioning toward network-based analysis for future planning cycles. Addressing the identified Triple Gap areas should be treated as an urgent planning priority to ensure equitable emergency service access for all residents of Karlsruhe.

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